

Deep Learning Assignment 1

# **Facial Affect Recognition**

22i-1959

# **Model Comparison Summary**

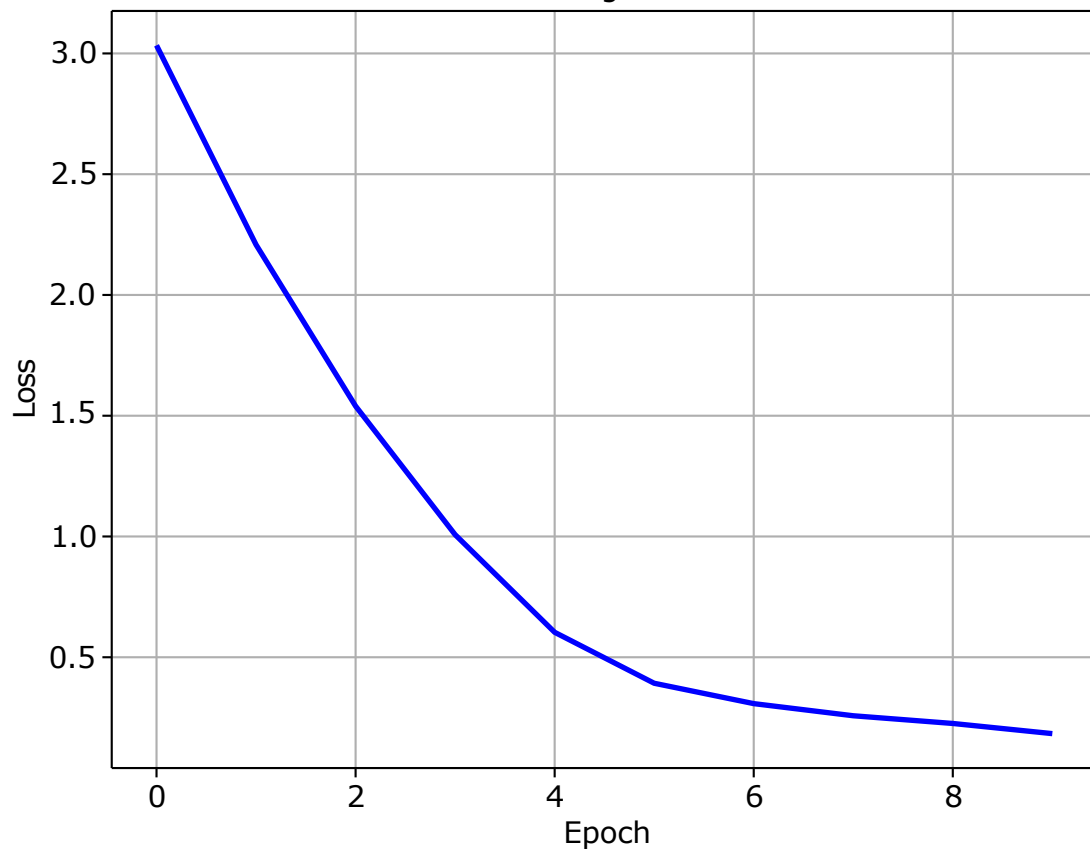
## Model: RESNET18

TEST METRICS (Final Results):

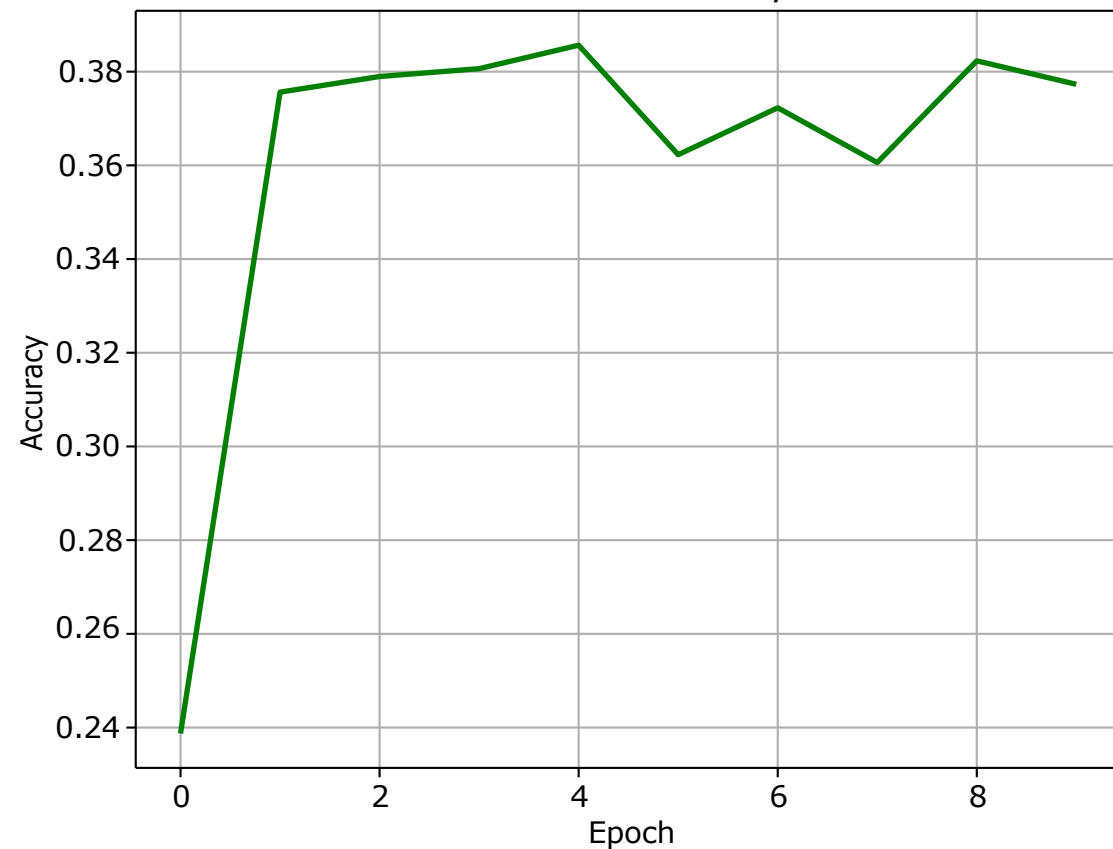
```
=====
acc                : 0.3823
f1_macro           : 0.3665
roc_auc            : 0.7593
pr_auc             : 0.3558
kappa              : 0.2929
krippendorff_alpha : 0.2930
val_rmse           : 0.4011
aro_rmse           : 0.3636
val_corr           : 0.5362
aro_corr           : 0.3518
val_sagr           : 0.7579
aro_sagr           : 0.7846
val_ccc            : 0.4984
aro_ccc            : 0.3161
```

# Training Progress - resnet18

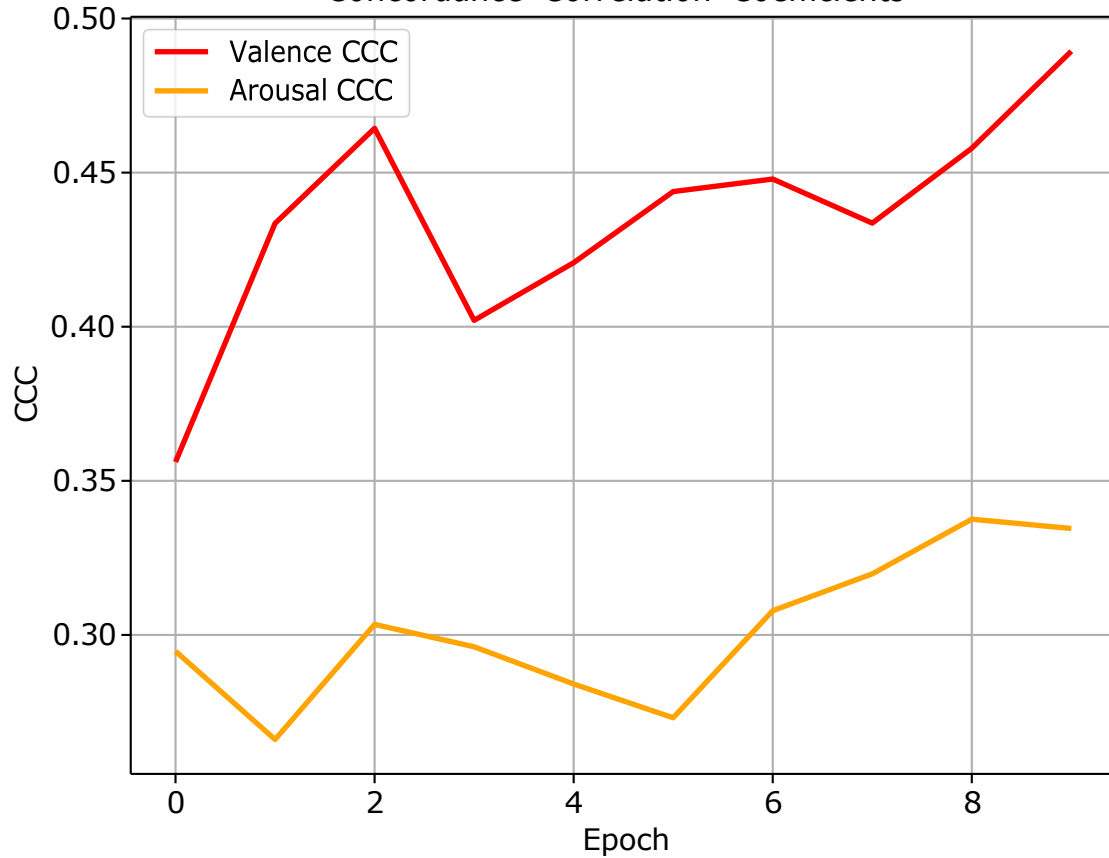
## Training Loss



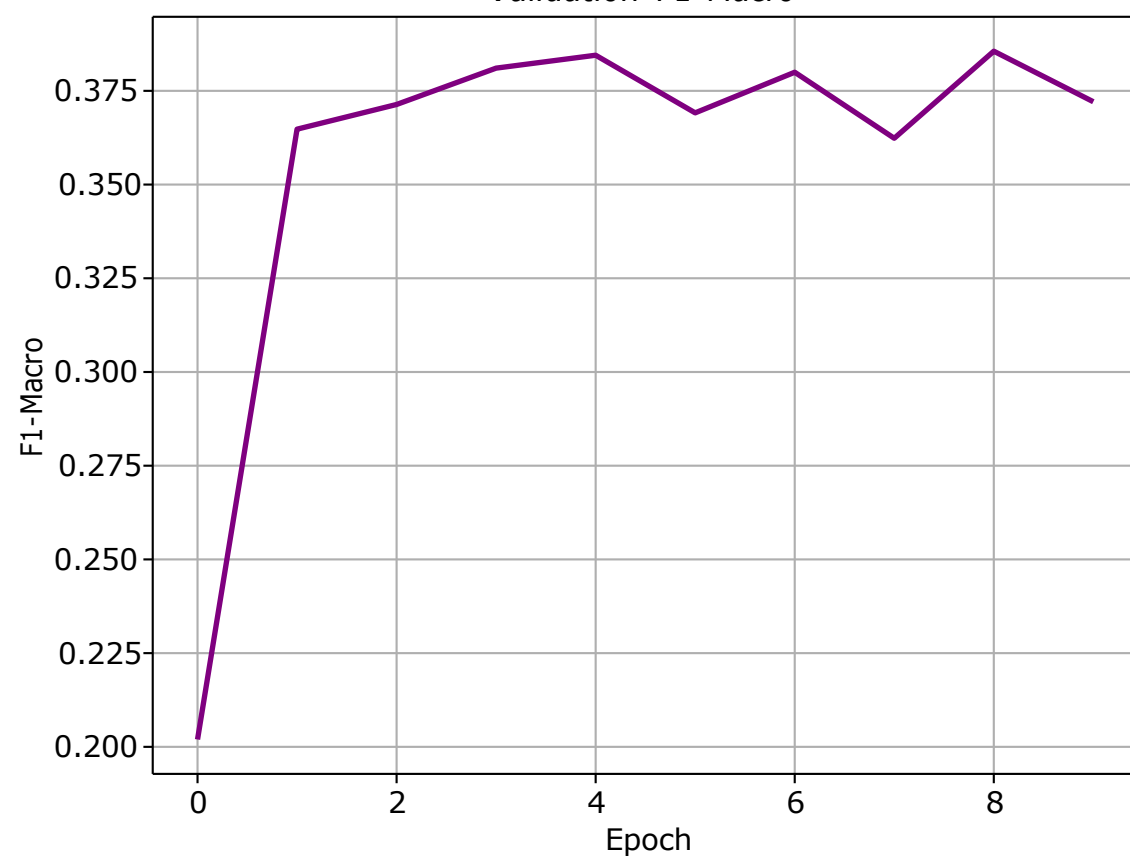
## Validation Accuracy



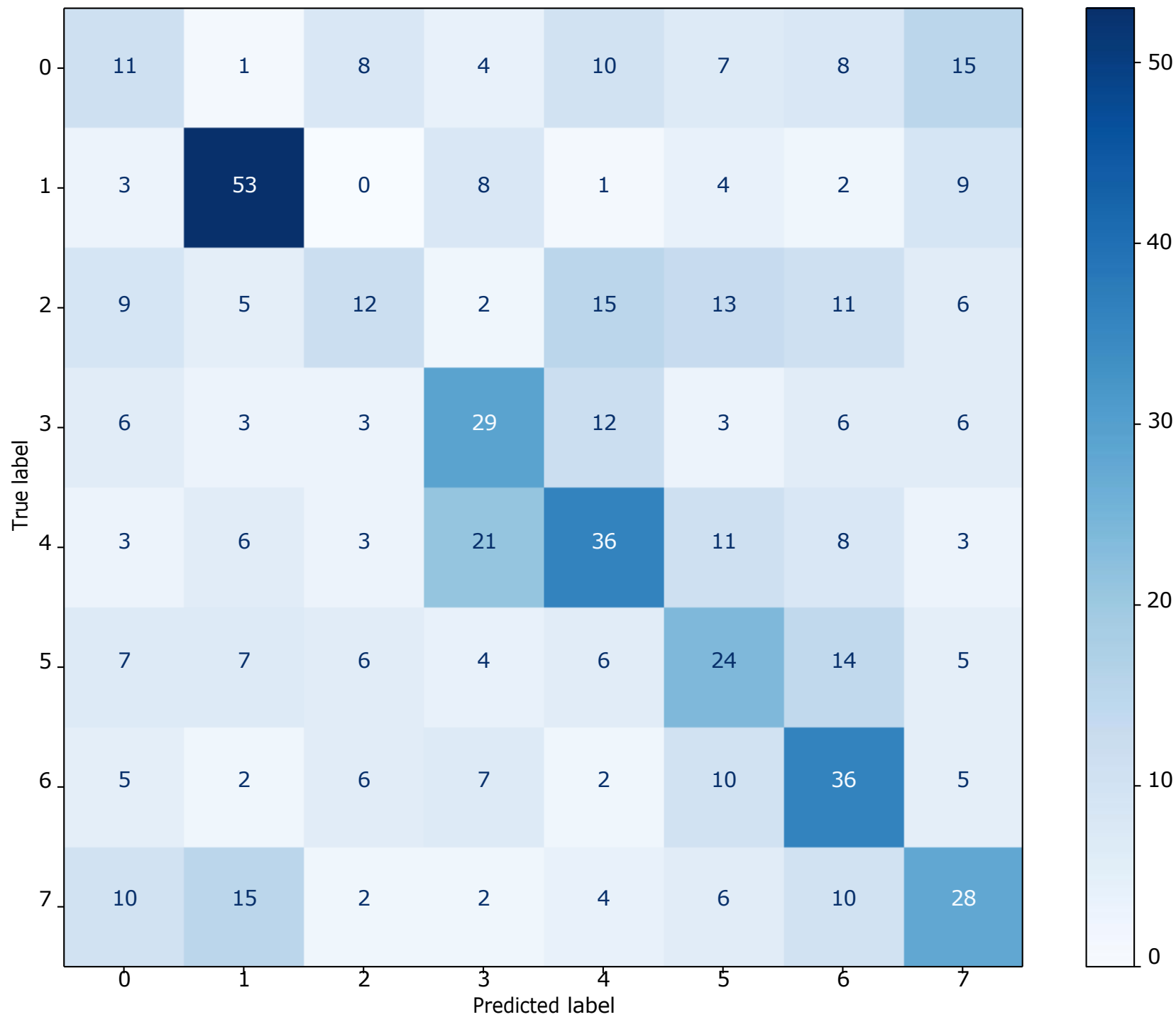
## Concordance Correlation Coefficients



## Validation F1-Macro

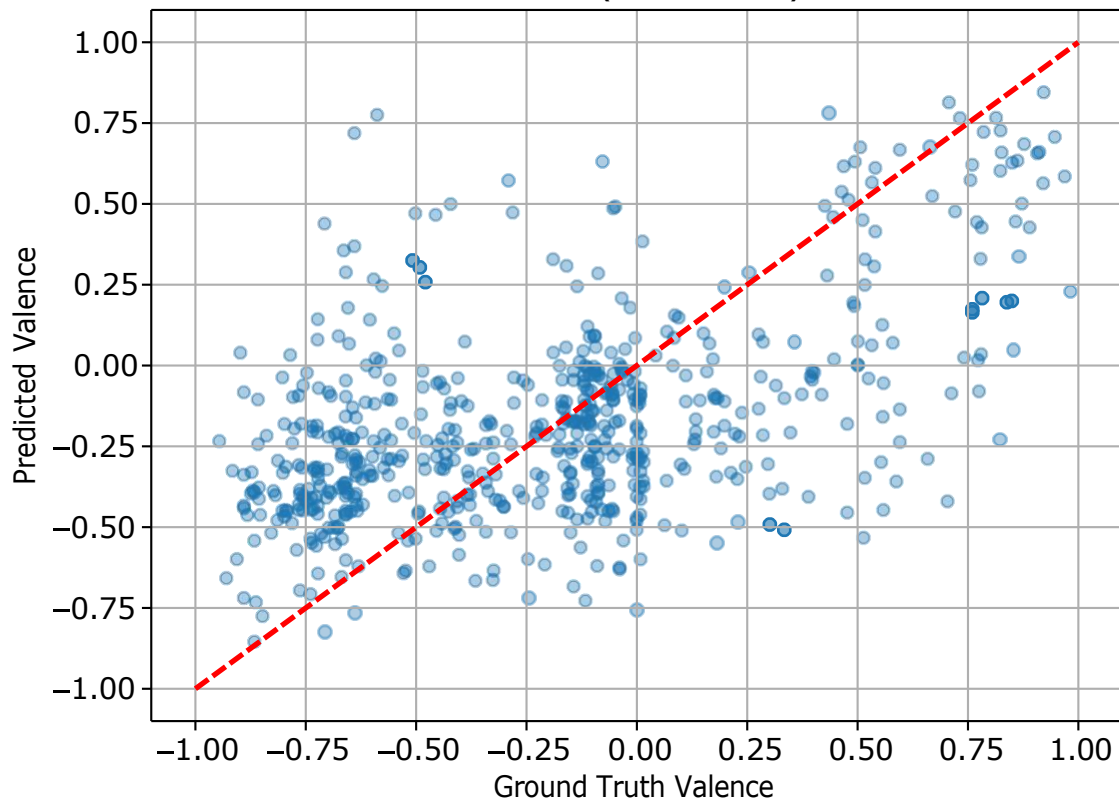


**Confusion Matrix - resnet18**

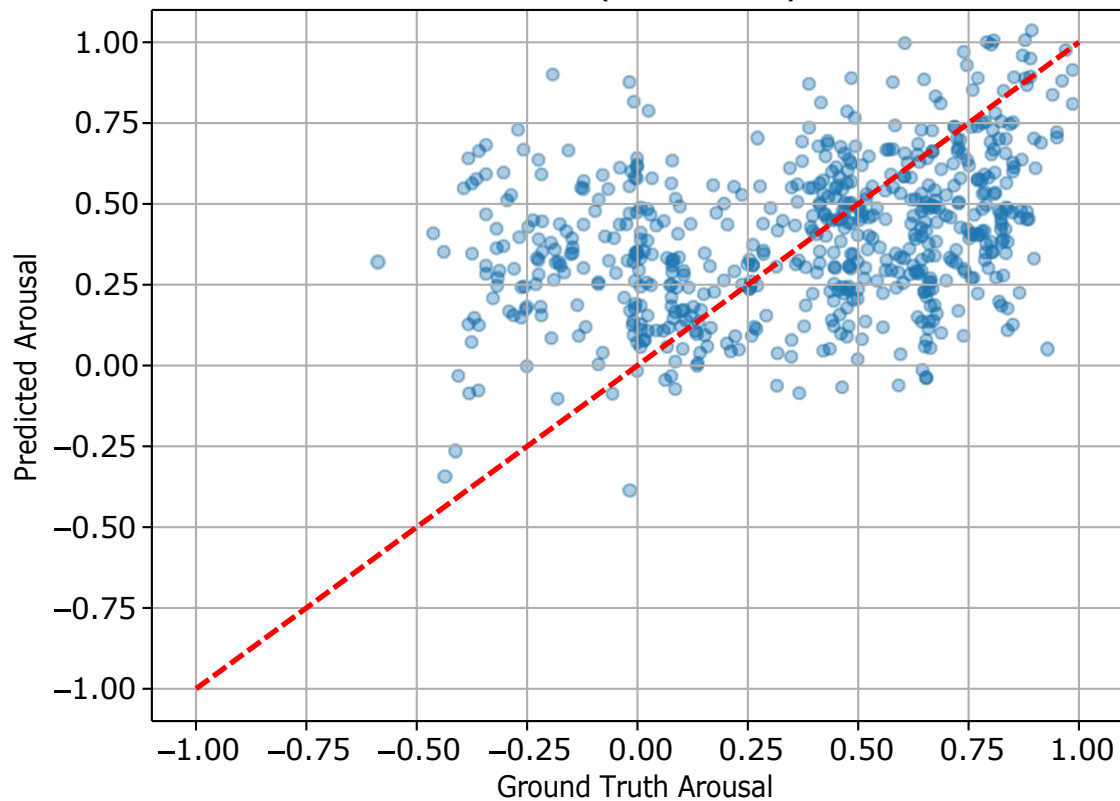


## Regression Results - resnet18

Valence (CCC: 0.498)



Arousal (CCC: 0.316)



## Correct Predictions - resnet18

ACT:4 Pred:4  
Val:-0.17 Aro:0.79



ACT:1 Pred:1  
Val:0.54 Aro:0.01



ACT:7 Pred:7  
Val:-0.54 Aro:0.67

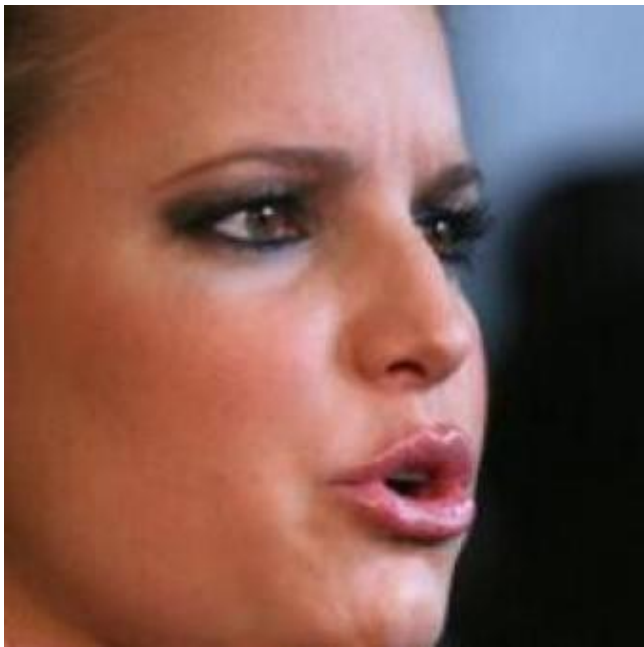


ACT:7 Pred:7  
Val:-0.64 Aro:0.67



## Incorrect Predictions - resnet18

ACT:6 Pred:3  
Val:-0.32 Aro:0.82



ACT:2 Pred:0  
Val:-0.63 Aro:-0.38



ACT:3 Pred:6  
Val:0.10 Aro:0.41



ACT:2 Pred:5  
Val:-0.68 Aro:-0.34





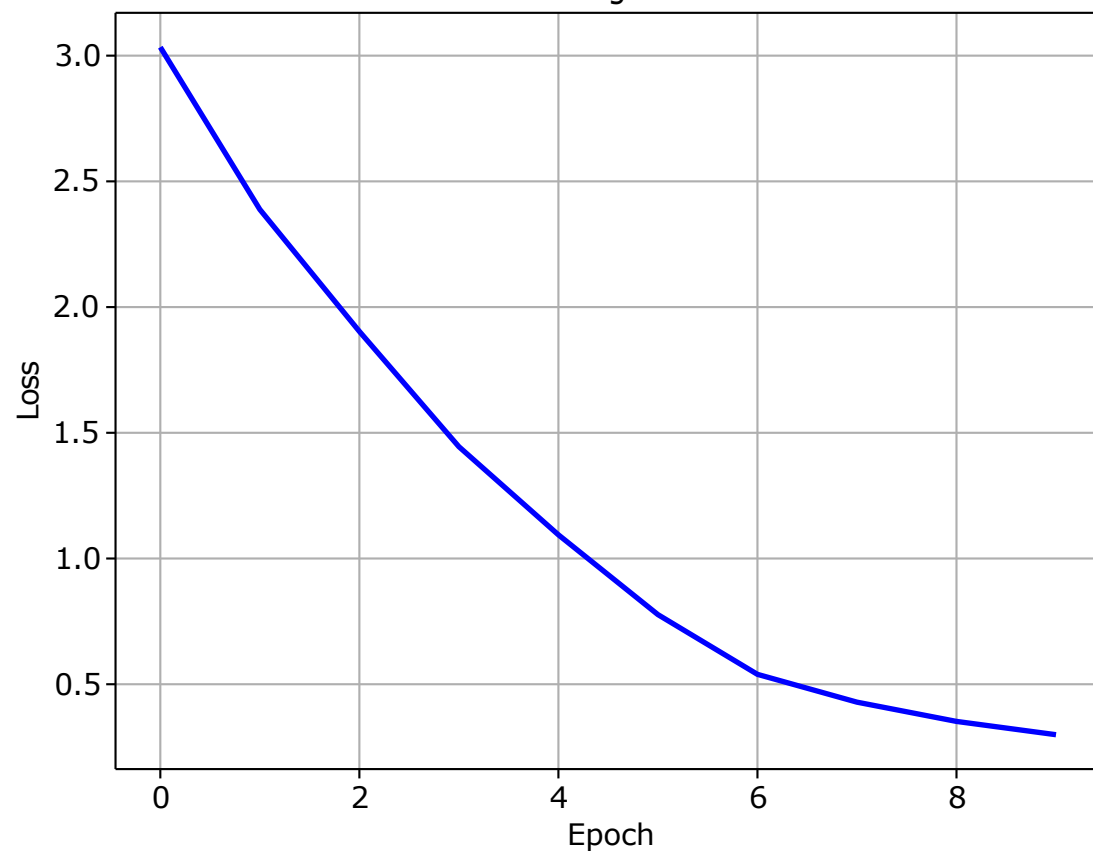
## Model: MOBILENET\_V2

TEST METRICS (Final Results):

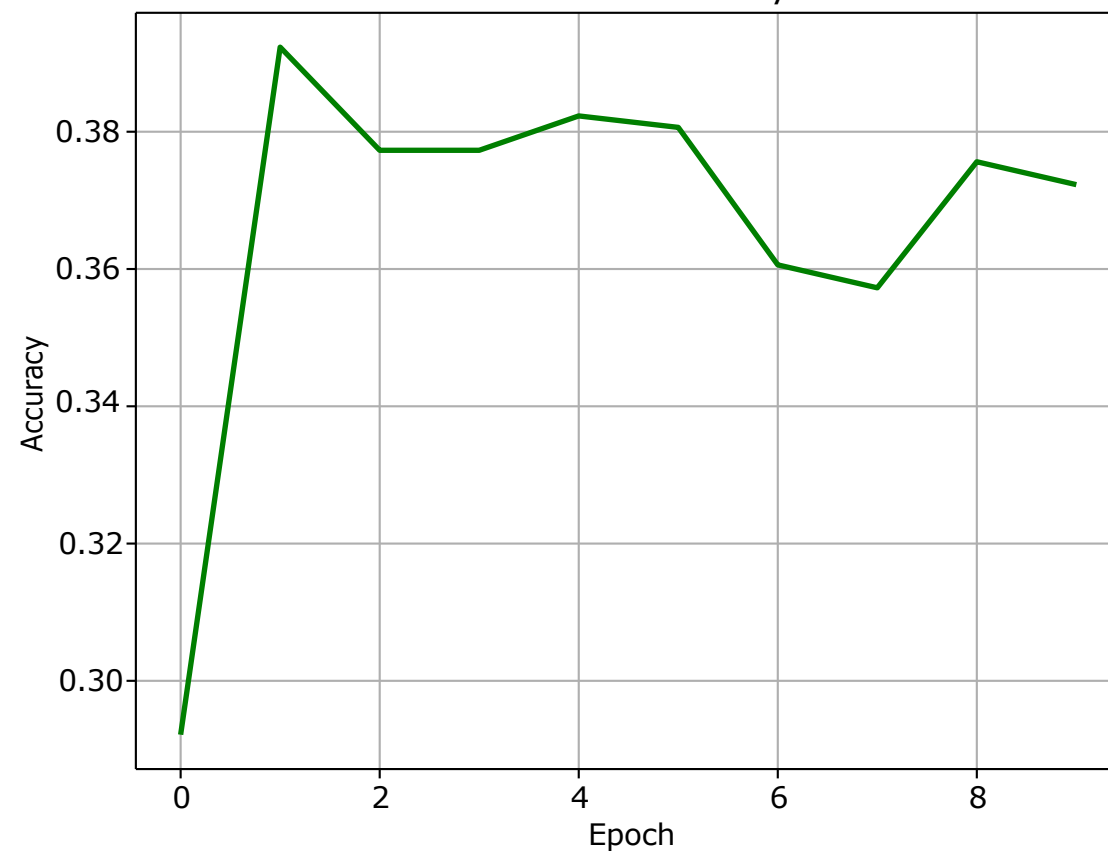
```
=====
acc                : 0.3556
f1_macro           : 0.3342
roc_auc            : 0.7798
pr_auc             : 0.3578
kappa              : 0.2643
krippendorff_alpha : 0.2624
val_rmse           : 0.4504
aro_rmse           : 0.4157
val_corr           : 0.4798
aro_corr           : 0.4000
val_sagr           : 0.6795
aro_sagr           : 0.7179
val_ccc            : 0.4635
aro_ccc            : 0.3938
```

# Training Progress - mobilenet\_v2

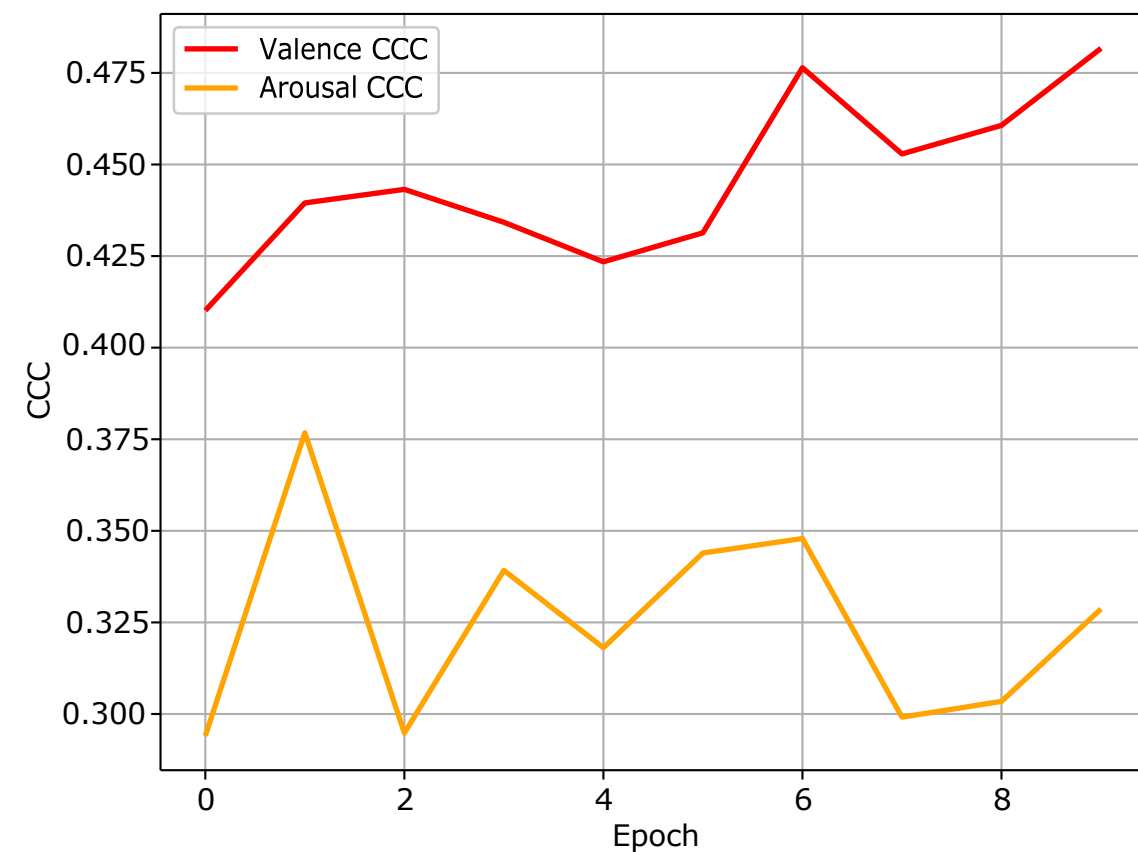
## Training Loss



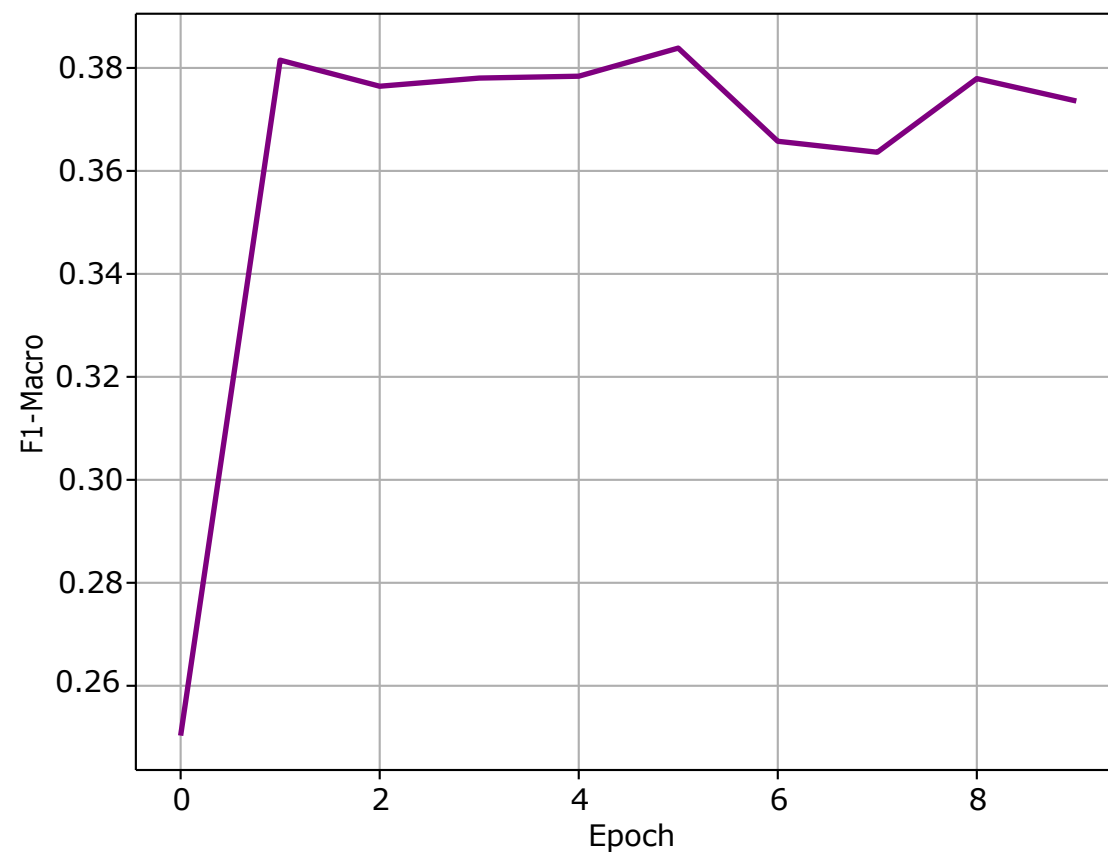
## Validation Accuracy



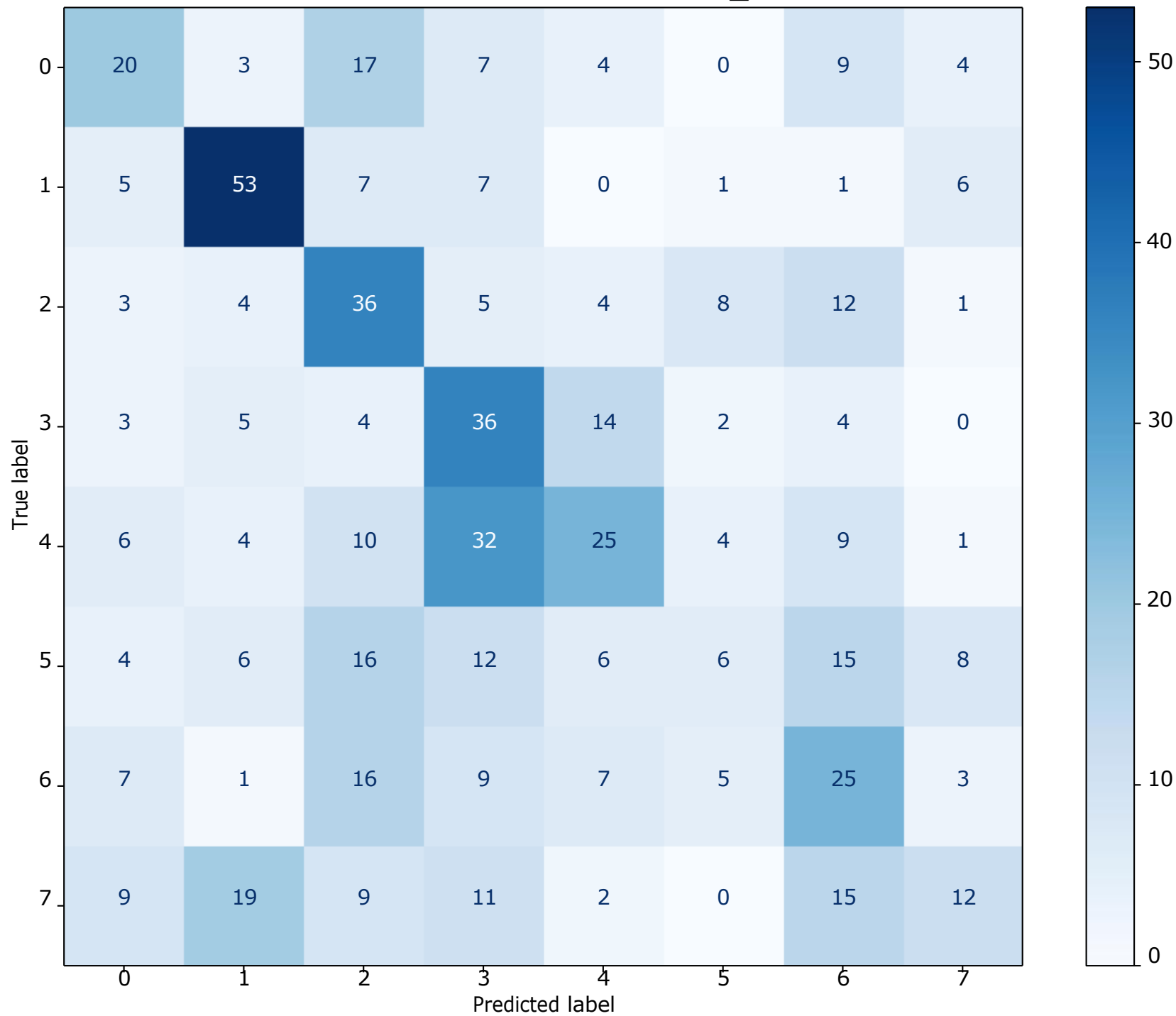
## Concordance Correlation Coefficients



## Validation F1-Macro

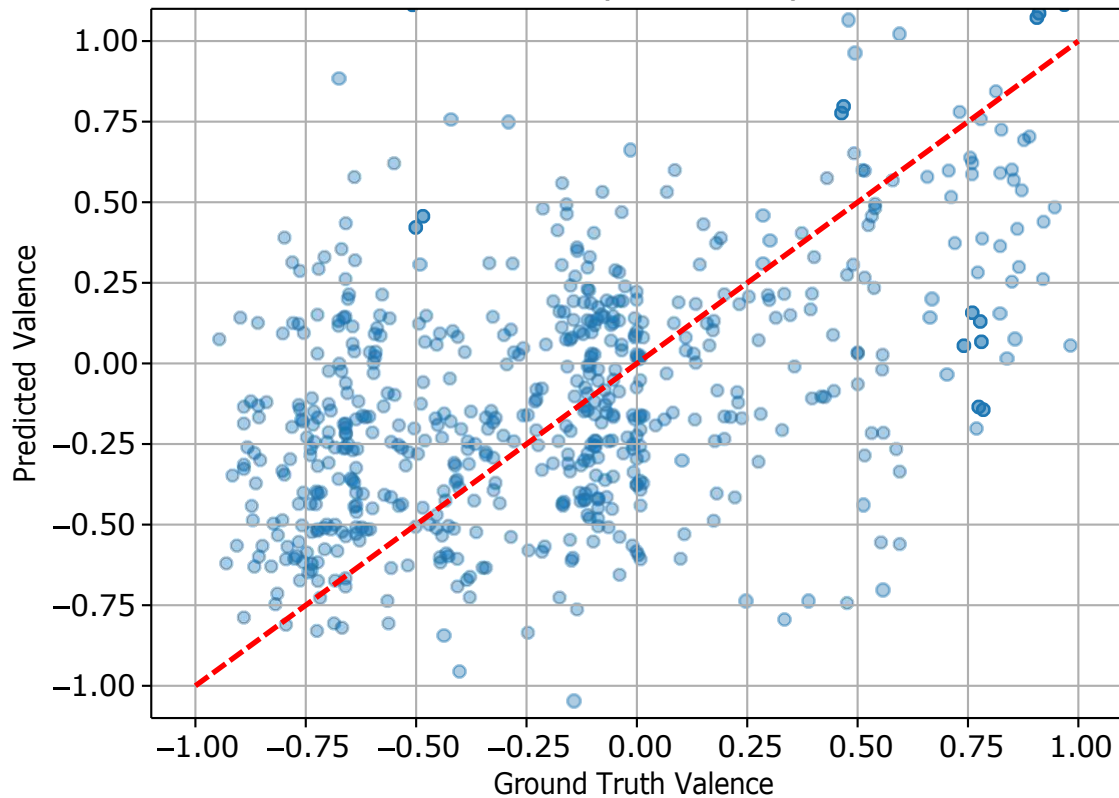


**Confusion Matrix - mobilenet\_v2**

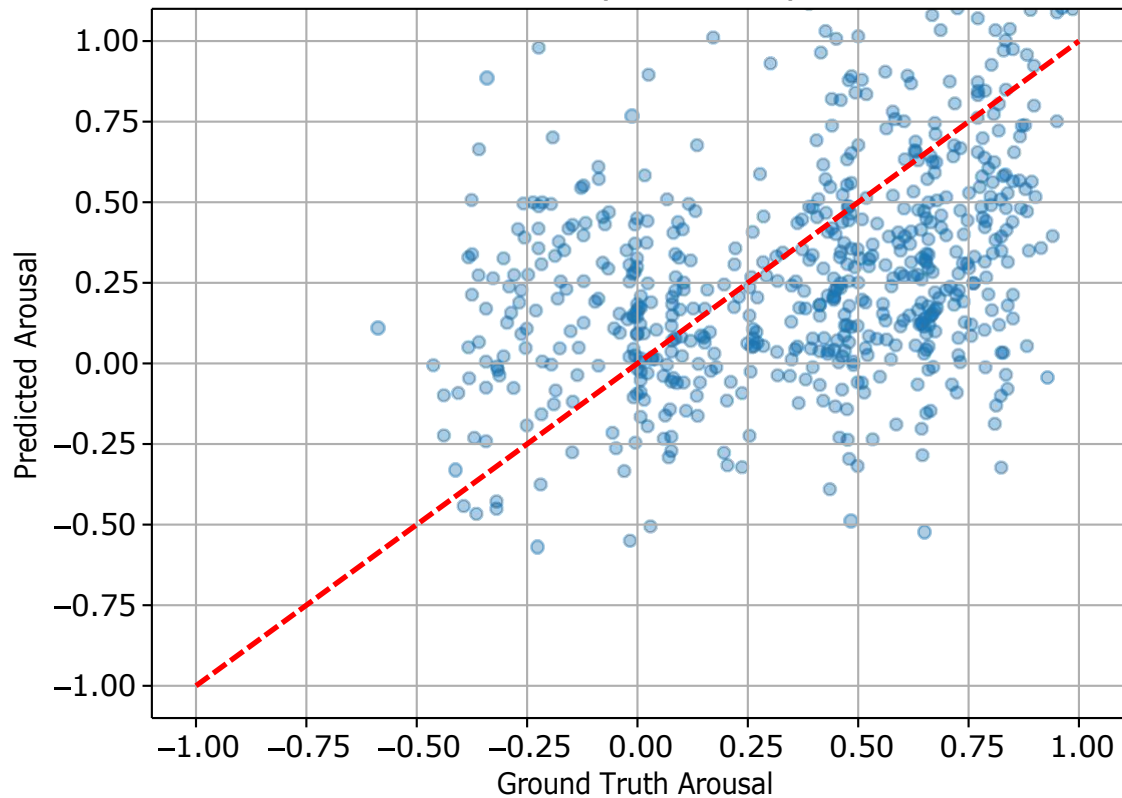


## Regression Results - mobilenet\_v2

Valence (CCC: 0.464)



Arousal (CCC: 0.394)



## Correct Predictions - mobilenet\_v2

GT:7 Pred:7  
Val:-0.22 Aro:0.24



T:4 Pred:4  
Val:-0.13 Aro:0.75



GT:2 Pred:2  
Val:-0.66 Aro:-0.18



GT:6 Pred:6  
Val:-0.56 Aro:0.25



# Incorrect Predictions - mobilenet\_v2

GT:7 Pred:1  
Val:-0.58 Aro:0.65



GT:5 Pred:1  
Val:-0.71 Aro:0.44



GT:7 Pred:3  
Val:-0.64 Aro:0.67



GT:3 Pred:0  
Val:0.08 Aro:0.27

